INCH-POUND

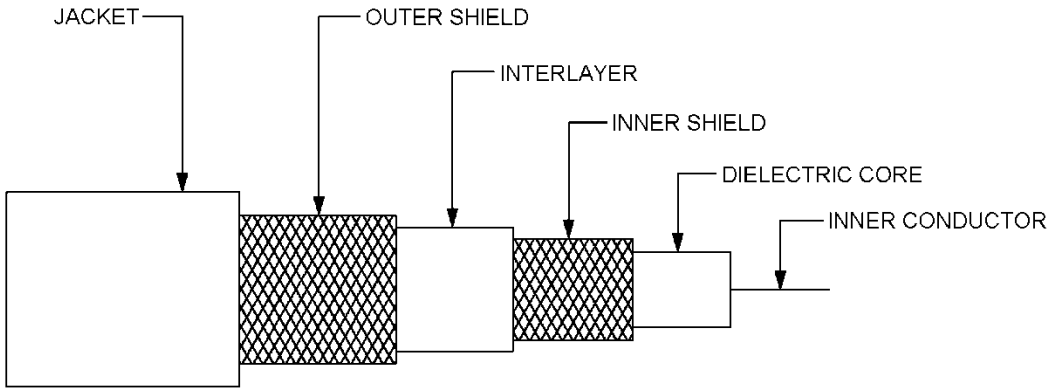
MIL-DTL-17/134D
w/AMENDMENT 2
18 March 2025
SUPERSEDING
MIL-DTL-17/134D
w/AMENDMENT 1
25 February 2015

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, TRIAXIAL, .245 INCH, 50 OHM, WATER BLOCKED, NON-WATER BLOCKED AND CROSSLINKED, LOW SMOKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.



NOTE: Cable should be constructed of non-toxic material.

FIGURE 1. General configuration.

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TABLE I. Description.

Component	Construction
Inner conductor	Solid strand of silver-coated copper wire Diameter: .033 inch ±0.001
Dielectric core	Type A-I: Solid polyethylene Diameter: .116 inch ± 0.004
Inner shield	Single braid of AVG No. 36 silver-coated copper wire Diameter: .150 inch maximum Coverage: 90%, minimum Carriers: 16 Ends: 6 Picks/inch: 10.4 Separator tape optional
Interlayer (M17/134-00001 and M17/134-00002)	Type A-1: Solid polyethylene Diameter: .175 inch ± .005
Interlayer (M17/134-00003 and M17/134-00004)	Cross-linked polyolefin Diameter: .175 inch ± .005
Outer shield M17/134-00001 through M17/134-00004	Single braid of AWG No. 36 silver-coated copper wire Diameter: .203 inch, maximum Coverage: 90%, minimum Carriers: 16 Ends: 8 Picks/inch: 9.8 Separator tape optional
Jacket M17/134-00003 through M17/134-00004	Cross-linked polyolefin Diameter: .245 inch, ± .005
Jacket (Inactive for new design) M17/134-00001 through M17/134-00002	Type IIIa: Polyethylene Diameter: .245 inch ± .005

NOTE: Free strippable blocking compound needed where necessary to meet hydrostatic test. The inner shield may be partially imbedded in the dielectric. Applicable to water blocked cables only.

ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

General configuration: See figure 1.

Power rating: See figure 2.

Operating temperature range:

-40°C to +70°C (M17/134-00001 and M17/134-00002)
-30°C to +85°C (M17/134-00003 and M17/134-00004)

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Inner conductor properties:

DC resistance (maximum at 20°C): 1.05 ohms per 100 feet.

Elongation: 25 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable useful for non-hosing low temperature applications.

REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

M17/134-0000X: 2 pounds minimum, 20 pounds maximum.

Aging stability:

+90°C ± 2°C (M17/134-00001and M17/134-00002)
+98°C ± 2°C (M17/134-00003 and M17/134-00004)

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend:

-55°C ± 2°C (M17/134-00001 and M17/134-00002)
-30°C ± 2°C (M17/134-00003 and M17/134-00004)

Dimensional stability: +85°C ± 2°C.

Inner conductor from core: .062 inch, maximum.

Inner conductor from jacket: .250 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Flame propagation: Applicable. Applicable to M17/ 134-00003 and M17/ 134-00004.

Acid gas generation: 2.0 percent, maximum. Applicable to M17/ 134-00003 and M17/ 134-00004.

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Halogen content: 0.2 percent, maximum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Immersion test: Applicable to M17/ 134-00003 and M17/ 134-00004.

Tensile strength, percent of unaged minimum: 50. Applicable to M17/ 134-00003 and M17/ 134-00004.

Elongation, percent of unaged minimum: 50. Applicable to M17/ 134-00003 and M17/ 134-00004.

Smoke index: 25 maximum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Toxicity index: 5 maximum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Durometer hardness: (type A) 80 minimum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Weathering: Applicable to M17/ 134-00003 and M17/ 134-00004.

Abrasion resistance: 75 cycles minimum (jacket only). Applicable to M17/ 134-00003 and M17/ 134-00004.

Tear strength: 35 pounds per inch minimum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Heat distortion: 30 percent maximum distortion. Applicable to M17/ 134-00003 and M17/ 134-00004.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Elongation: 160 percent, minimum. Applicable to M17/ 134-00003 and M17/ 134-00004.

Physical tests on aged jacket: Applicable to M17/ 134-00003 and M17/ 134-00004.

Air oven:

Tensile strength, percent minimum: 60.

Elongation, percent minimum: 60.

Hot oil immersion: Applicable to M17/ 134-00003 and M17/ 134-00004.

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent, minimum. Applicable to M17/ 134-00003 and M17/ 134-00004.

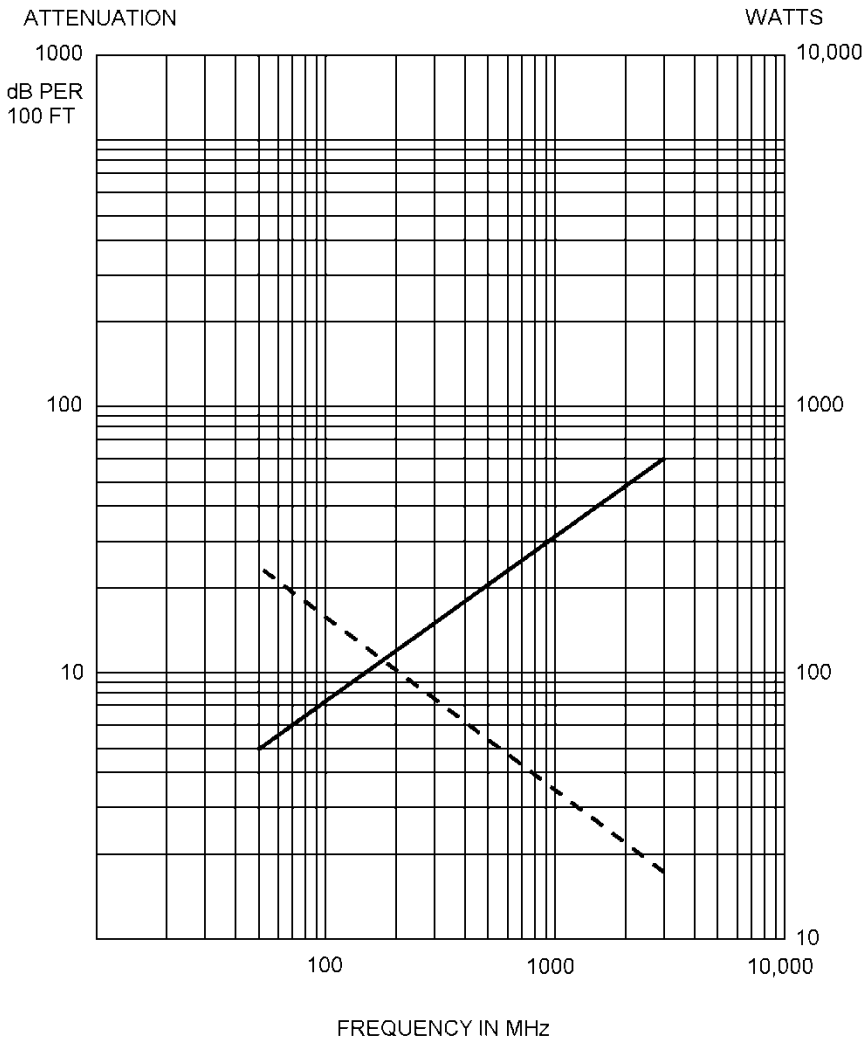
Weight: 4.5 pounds per 100 feet, max (M17/134-00001 and –00002).
5 pounds per 100 feet, max (M17/134-00003 and –00004).

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Watertightness (not applicable to M17/134-00002 or M17/134-00004): One end of a five foot length of completed cable shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 lb_f/in^2 for a period of 6 hours. Any evidence of water leakage shall be cause for rejection.

Hydrostatic (open end) (not applicable to M17/134-00002 or M17/134-00004): A 5-foot specimen of completed cable shall be fitted into a gland similar to USL 44500B-12. The cable shall be capable of withstanding 1,000 lb_f/in^2 applied to the gland end of the specimen for a period of 2 hours. Any evidence of water leakage shall be cause for rejection.

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Maximum power (at 25°C sea level) - - - - - - - - - - - - - -

Maximum attenuation————————

Frequency MHz	Attenuation dB	Watts dB
50	4	240
100	6	150
1000	26	33
3000	60	16

Test requirements shall be noted as line indicated on graph.

FIGURE 2. Power rating and attenuation.

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SWR	Reflection coefficient	Return loss dB	SWR	Reflection coefficient	Return loss dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

Frequency MHz	MIN. SRL
50	27
400	24.4
1000	22.5
2000	21.5
3000	20

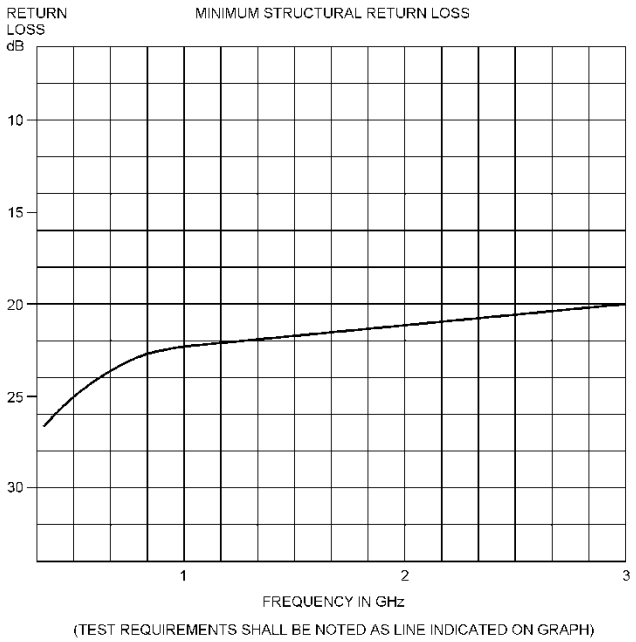


FIGURE 3. Structural return loss.

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Electrical:

Continuity: Applicable.

Spark test: 5,000 V rms, +10%, -0%.

Voltage withstanding: 5,000 V rms, minimum applied between the inner and outer conductors with the outer conductor grounded; 500 V dc minimum applied between the outer conductor and the extra shield with the extra shield grounded.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Marking: Applicable.

Marking Durability: 125 cycles (250 strokes), 1 pound.

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TABLE I. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				

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TABLE I. Qualification and Group A, B, and C Table – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C
	Weight	3.9	4.8.24	1	Per Table VIII	B

Part or Identifying Number (PIN): See table II.

Supersession data: See table II.

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TABLE II. Cross-reference of PIN.

PIN	Superseded PIN or type designation	Type
M17/134—00001 1/	TRF58	Water blocked
M17/134—00002 1/	YR19210	Non-water blocked
M17/134—00003 2/	AA-7557	Water blocked
M17/134—00004 2/	AA-7558	Non-water blocked

1/ Inactive for new design.
2/ Cable previously manufactured using the barrier tape is acceptable for Government use until stock is purged.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity
DLA - CC

(Project 6145-2025-002)

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19
NASA - NA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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